Clockwork Tyres Roadmap

Date: March 29, 2026

1. Executive Summary

Clockwork Tyres should be built as a unified platform with three operating modes:

- Super Admin
- Supplier Admin
- Retailer Admin plus Retail Storefront

Recommended system split:

- `reporting-crm` becomes the core platform and system of record.
- `dealer-portal-Angular` becomes the reference for the retail/storefront experience.
- `tunerstop-vendor` becomes reference-only for old business rules, plan logic, and migration mapping.

This is not a skinning exercise. It is a structured platform rebuild on top of the new CRM, with selective frontend reuse.

2. Target Product Vision

Super Admin

Platform owner view for:

- all suppliers
- all retailers and subscribers
- plan and subscription oversight
- wholesale enablement
- account status and lifecycle
- platform analytics
- error monitoring and operational visibility

Supplier Admin

Supplier portal based on the CRM for:

- products
- product images
- inventory
- warehouses
- quotes and proformas
- invoices
- customers
- users and permissions
- analytics
- storefront preview of only that supplier's products

Retailer Admin plus Storefront

Retailer gets:

- CRM admin portal
- retail-facing storefront for in-store or customer-assisted sales
- customer capture during checkout
- invoice creation from storefront orders
- supplier exploration inside admin
- supplier connection requests and approvals
- procurement flow inside admin
- analytics for retail operations

Core business shift:

- B2B supplier discovery and procurement move into admin.
- The storefront becomes retail-facing.

3. Architecture Decision

Base Platform

Use `reporting-crm` as the new base.

Reasons:

- it already has roles and permissions
- it already has quotes, invoices, inventory, warehouses, reports, and customer modules
- it already has a storefront API layer
- it already separates admin users from dealer/storefront users

Legacy Platform

Use `tunerstop-vendor` only for:

- supplier-retailer relationship reference
- old plan and entitlement logic
- old onboarding and invite behavior
- migration mapping
- analytics/reporting reference

Do not continue building on `tunerstop-vendor` as the future runtime backend.

Storefront

Use the current Clockwork frontend as reference and partial code-reuse source, but do not keep the old API contract forever.

4. Required Core Platform Additions

Before the new vision can work cleanly, the CRM needs these new foundations:

- `accounts` or `organizations`
- `account_users`
- `account_types`
- `subscriptions`
- `plan_features`
- `supplier_relationships`
- account-scoped branding and settings
- account-scoped products, inventory, warehouses, customers, and orders

This is the most important missing layer.

Without account tenancy, supplier data isolation, retailer permissions, and super-admin reporting will become messy.

5. Repository Strategy

Backend Repo

Current repo:

- `clockwork-tyres`

Recommendation:

- This name is acceptable if this repo is the main platform repo.
- If you want a clearer name before the project grows, rename it early to one of:
  - `clockwork-tyres-platform`
  - `clockwork-tyres-core`
  - `clockwork-tyres-backend`

Best choice:

- `clockwork-tyres-platform`

Reason:

- the repo will hold more than a backend
- it is the platform core for super admin, supplier admin, and retailer admin

Frontend Repo

Recommended:

- create a new frontend repo

Suggested names:

- `clockwork-tyres-storefront`
- `clockwork-tyres-web`

Best choice:

- `clockwork-tyres-storefront`

Reason:

- the current Angular frontend is tightly coupled to old wheel-focused APIs and old Clockwork assumptions
- a new repo gives clean boundaries, clearer release management, and less accidental reuse of legacy behavior
- the old frontend repo can stay as a reference implementation

Legacy Repos

Keep these as reference-only:

- `dealer-portal-Angular`
- `tunerstop-vendor`

Do not merge them into the new repo. Use them for guided extraction and migration.

6. Delivery Principles

- Build on the new CRM, not on the legacy backend.
- Separate account ownership from user role permissions.
- Freeze the tire catalog schema before major UI work.
- Move supplier discovery and procurement into admin.
- Keep rollout phased and pilot-driven.
- Reuse proven flows where possible, but do not preserve bad architecture for speed.

7. Workstreams

The roadmap should run across five workstreams:

- Platform and tenancy
- Catalog and inventory
- Admin experiences
- Storefront experience
- Migration, QA, and rollout

8. Phase-by-Phase Roadmap

Phase 0: Discovery and Scope Lock

Goal

Freeze the business model and technical direction before implementation.

Key Decisions To Lock

- business/account types: supplier, retailer, both
- subscription matrix
- supplier count limits
- tire product schema
- service items and pricing behavior
- retail pricing rules
- procurement workflow
- payment model
- migration scope
- reporting requirements

Tasks

- convert George's notes into a formal product brief
- produce a roles and permission matrix
- produce a supplier-retailer lifecycle diagram
- define tire catalog fields and import format
- define procurement lifecycle from retailer invoice to supplier approval
- define subscription entitlements by plan
- define data migration scope for accounts, products, inventory, and orders

Deliverables

- approved product brief
- approved account model
- approved subscription matrix
- approved tire data schema
- approved procurement flow

- approved migration scope

Exit Criteria

- no major business ambiguity remains around plans, roles, or catalog structure

Estimate

- 1 to 2 weeks

Phase 1: Platform Foundation in CRM

Goal

Turn `reporting-crm` into a multi-account platform foundation.

Scope

- account tenancy
- user-to-account membership
- plan and subscription foundation
- account-level settings
- account scoping in admin resources

Tasks

- add `accounts` table
- add `account_users` table
- add account type enum or model
- add subscription and plan tables
- add account branding and feature flags
- attach ownership to users, warehouses, products, inventory, customers, and orders
- add global super-admin bypass
- add account-aware policies and query scopes
- add audit logging for account actions

Deliverables

- CRM supports multiple supplier and retailer businesses in one platform

- admin resources are account-aware
- platform can distinguish internal roles from business entitlements

Exit Criteria

- supplier and retailer data are isolated cleanly
- super admin can view all accounts
- non-super-admin users only see their own account data

Estimate

- 2 to 3 weeks

Phase 2: Tire Catalog and Inventory Model

Goal

Replace wheel-oriented assumptions with a tire-first catalog.

Scope

- products
- variants
- filters
- fitment/search inputs
- imports and exports
- inventory representation

Suggested Tire Fields

- brand
- model
- section width
- aspect ratio
- rim size
- load index
- speed rating
- season
- run flat

- XL or reinforced
- origin
- year
- SKU
- supplier cost
- retail price
- track inventory
- available quantity by warehouse

Tasks

- redesign product and variant schema
- redesign CSV import and export structure
- update product grid and inventory grid
- update search/filter service layer
- add service items such as mounting and balance, nitrogen, and wheel alignment
- define image strategy for tire products

Deliverables

- tire-ready catalog
- tire-ready import pipeline
- updated inventory model
- updated reporting dimensions

Exit Criteria

- one supplier can manage a complete tire catalog and inventory in CRM

Estimate

- 2 to 3 weeks

Phase 3: Super Admin Module

Goal

Build the platform owner view shown in George's super-admin mockups.

Scope

- all accounts
- subscription overview
- activation and wholesale enablement
- top-level analytics
- platform health and error log

Tasks

- build account listing with status, type, and subscription
- show retailer count, supplier count, subscriber count
- add wholesale-enabled flag
- add account-level analytics cards
- add error log visibility
- add actions for activate, suspend, and manage plan

Deliverables

- super-admin dashboard
- accounts table
- subscription management screens
- operational reporting

Exit Criteria

- platform operators can fully manage supplier and retailer businesses

Estimate

- 1 to 2 weeks

Phase 4: Supplier Admin Module

Goal

Adapt CRM into a supplier-facing admin portal.

Scope

- products and product images
- inventory and warehouses
- quotes and proformas
- invoices
- customers
- users
- analytics
- storefront preview

Tasks

- scope all CRM resources to supplier account
- adapt labels and flows from reporting to tires
- add supplier users and permission templates
- add storefront preview mode
- add notifications for new procurement requests
- verify warehouse and shipping behavior

Deliverables

- supplier admin portal
- supplier-scoped catalog and inventory
- supplier preview storefront mode

Exit Criteria

- one supplier can fully run their operations from CRM without legacy backend

Estimate

- 2 to 4 weeks

Phase 5: Retailer Admin and Supplier Procurement

Goal

Build the retailer-side admin behavior that sits behind the storefront.

Scope

- supplier discovery
- supplier connection requests
- connected supplier management
- procurement search
- procurement cart
- procurement request submission
- supplier-side approval path

Tasks

- add supplier relationship model
- add request, approve, reject flow
- add explore suppliers screen in admin
- add my suppliers screen in admin
- build procurement search by size and supplier
- build procurement cart
- link retailer invoice to "Procure Tires"
- generate supplier-side quote or proforma from procurement request
- notify supplier by email and in-app

Deliverables

- retailer admin for supplier connections
- retailer procurement workflow
- supplier-side procurement intake

Exit Criteria

- a retailer can source products from approved suppliers entirely inside CRM

Estimate

- 3 to 4 weeks

Phase 6: Storefront Build

Goal

Create the new retail-facing Clockwork storefront as a separate repo and connect it to CRM APIs.

Scope

- authentication
- vehicle search
- size search
- catalog listing
- product detail
- cart
- checkout
- account area
- order history
- supplier preview mode

Tasks

- create `clockwork-tyres-storefront`
- port only the useful Angular UI patterns from old Clockwork
- remove public supplier discovery from storefront
- update branding and tire-focused UI
- connect login and account flows to CRM APIs
- connect search and catalog to new tire APIs
- connect cart and checkout to retail order flow
- create supplier preview mode with no cart or checkout

Deliverables

- production storefront repo
- retailer sales flow
- supplier preview mode

Exit Criteria

- retailer can use storefront to create customer sales

- supplier can preview their own catalog safely

Estimate

- 3 to 5 weeks

Phase 7: Integration and Data Migration

Goal

Move from reference systems into the new platform safely.

Scope

- account migration
- user migration
- supplier-retailer relationship migration
- plan migration
- catalog and inventory migration
- order history migration if approved

Tasks

- map legacy dealers to retailer accounts
- map legacy vendors to supplier accounts
- map `dealer_vendors` to new supplier relationships
- map legacy plans to new subscriptions
- import products and inventory according to final tire schema
- migrate selected orders and invoices if required
- run reconciliation and data validation scripts

Deliverables

- migration scripts
- migration runbook
- validation reports

Exit Criteria

- pilot accounts can work only from new platform without depending on legacy backend

Estimate

- 2 to 4 weeks

Phase 8: QA, Pilot, and Launch

Goal

Harden the platform and release gradually.

Scope

- QA
- UAT
- pilot rollout
- training
- launch

Test Coverage Required

- self-registration
- invite-based onboarding
- supplier approval
- supplier preview mode
- retail checkout
- invoice creation
- procure tires flow
- supplier quote receipt
- quote to invoice conversion
- stock movement and updates
- subscription gating
- role and permission visibility

Tasks

- add contract tests for storefront APIs
- add end-to-end flows for key journeys
- run pilot with one supplier and one retailer
- fix defects
- train operations team
- enable phased rollout

Deliverables

- QA report
- UAT signoff
- pilot signoff
- launch checklist

Exit Criteria

- agreed pilot accounts operate successfully in production

Estimate

- 2 to 3 weeks

9. Suggested Timeline Summary

Phase	Name	Estimate
0	Discovery and scope lock	1 to 2 weeks
1	Platform foundation in CRM	2 to 3 weeks
2	Tire catalog and inventory model	2 to 3 weeks
3	Super admin module	1 to 2 weeks
4	Supplier admin module	2 to 4 weeks
5	Retailer admin and procurement	3 to 4 weeks
6	Storefront build	3 to 5 weeks

Phase	Name	Estimate
7	Integration and migration	2 to 4 weeks
8	QA, pilot, and launch	2 to 3 weeks

Realistic program range:

- 18 to 30 weeks depending on team size, scope lock, and migration depth

10. Recommended Team Structure

- Product and business analyst
- CRM backend lead
- Storefront frontend lead
- Full-stack integration developer
- QA lead
- DevOps and release owner

11. Open Questions For George

These must be answered before Phase 1 starts:

- Is each business one account that can be retailer, supplier, or both?
- Is supplier the same as wholesaler in all cases?
- What exactly is included in free versus paid plans?
- Is the three-supplier free limit final?
- Is the storefront public or counter-use only?
- Should end customers see supplier stock at all?
- Should procurement always start from an invoice, or can it be standalone?
- Should supplier approval reserve stock immediately?
- Are payments online, invoice-based, in-store, or mixed?
- What is the final tire import file and field list?
- How much history must be migrated?
- Do multi-branch retailers need multiple warehouses and staff users from day one?

12. Immediate Next Steps

Recommended next actions:

1. approve repo naming strategy
2. create the separate storefront repo
3. convert this roadmap into epics and tickets
4. finalize Phase 0 decisions with George
5. start Phase 1 only after tire schema and subscription rules are approved

13. Final Recommendation

Yes, create a new frontend repo.

Recommended setup:

- Backend/platform repo: `clockwork-tyres-platform`
- Frontend/storefront repo: `clockwork-tyres-storefront`

If you prefer not to rename today, keep:

- `clockwork-tyres` for platform/backend

and create:

- `clockwork-tyres-storefront` for frontend

That is the cleanest structure for long-term delivery.

14. What We Can Start Before George Replies

We do not need to wait for every open question before starting work.

Safe To Start Now

- repo bootstrap and CI
- backend module skeleton in `clockwork-tyres`
- account and tenancy foundation
- account memberships and capability flags
- route and API boundary design

- super-admin account shell
- billing and entitlement engine shell
- migration mapping from legacy systems
- API contract inventory from old frontend
- new storefront repo structure
- design system and route structure for storefront
- QA baseline, test strategy, and automation setup

Must Wait For George

- final tire import schema
- final subscription matrix
- exact account-type rules for supplier, retailer, and both
- procurement lifecycle state rules
- supplier stock visibility rules in storefront
- payment model for checkout
- final reporting and KPI definitions
- full migration depth for historical data

Execution Rule

Start platform structure, contracts, and low-regret foundations immediately.

Do not lock irreversible product schema, subscription behavior, or procurement workflows until the blocked decisions are answered.